

Inside The Mind of Artificial Intelligence

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ABSTRACT

This paper presents a comprehensive and accessible overview of modern Artificial Intelligence (AI)... (13), focusing on how contemporary AI systems function, learn, and generate content. It explains the foundational role of neural networks as the backbone of widely used models such as ChatGPT, image generation systems like Stable Diffusion, and scientific breakthroughs such as Alpha Fold. Through simplified examples, the paper describes supervised and unsupervised learning, reinforcement learning from human feedback (RLHF), and major neural network architectures including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers.

The study also addresses common misconceptions and societal concerns surrounding AI, particularly claims that AI systems copy or steal artistic and journalistic content. By comparing AI learning mechanisms with human learning and creative processes, the paper argues that AI primarily learns patterns and styles rather than memorizing or reproducing exact works. Legal and ethical debates, such as lawsuits from publishers, are examined through the lens of pattern recognition and information abstraction.

Furthermore, the paper evaluates claims regarding AI's ability to solve "unsolvable" problems, including cryptography and complex scientific challenges. While clarifying that AI cannot break modern encryption through shortcuts, it highlights AI's success in domains like protein folding, where traditional mathematical approaches have failed. Finally, the paper discusses whether AI can surpass human performance across domains, concluding that while AI excels at pattern-based tasks, its capabilities remain bounded by data, architecture, and human-defined objectives.

Overall, this paper aims to demystify AI by presenting its strengths, limitations, and implications in clear and non-technical terms.

1. INTRODUCTION

This one article is going to solve all the questions for you.

How dose AI work?

How dose AI learn?

How dose ChatGPT work?

How dose image generation work?

Dose AI copy/steal art or content?

I know a decent portion of artists that do not like Artificial Intelligence. Some of them quite hostile towards the AI because they think that AI is stealing their work or their Art styles and other groups that does not like AI very much or for example publishers not saying all of them but some of them like New York Times for example, they claim that OpenAI is coping their content and they are now suing OpenAI for this.

But is this really the case or a valid argument?

Also, Can AI solve unsolvable math problems?

For example, there is a myth that one team trained ai for braking encryption system. The system that secures our passwords, our bank accounts, the internet, government data etc. Now as far as we know that is no Mathematical viable way to rally hack this systematically. The only way is to brute force guess all the possibilities of pass words.

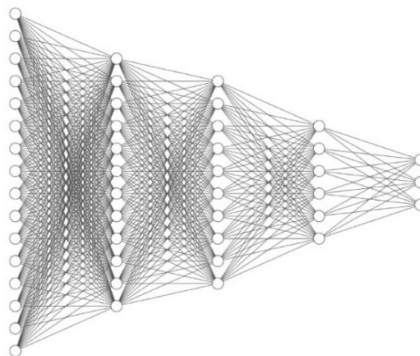
Can Ai really do this?

Also, Can Ai beat humans in everything?

Can Ai eventually be so good that it can outperforms humans at any tasks. We cover all of this in very easy terms.

The essence behind all AI system we know today Whether it's ChatGPT or Mid-journey or stable diffusion or Sora or AlphaFold. The backbone of all of these AI systems is the Nural Network... (1)(2).

Nural networks is looks like the this:



It's basically Layers of nodes so each circle in figure is called node and each line of nodes is called layers and each node is interconnected with one and other through the linkages... (1).

The Neuron network is designed based on human brain except for nodes and linkages in human brain ... (3). It's just a network of neurons and synapses.

2. How exactly an AI work?

Let me give you the simple example. Let's say we have a Neuron Network which is trained to identify the images of cats versus dogs. So, let's say we input/feed this neuron network with an image of cat. This image would actually be broken down in to data and the data will flow through each of these nodes and after these flows through the first layer of nodes, it will flow through the second layer of nodes and then the next layer of nodes and then the next layer and so on and so forth until it reaches the final layer. In which case it would calculate the values of this and one the based on the values of the final layer it would split out an answer "this is a cat".

In simple words, each node is a chef in a restaurant and each chef is assigned to add particular spices in the dish, like one is allocated for adding salt, one is for chilly, one for the oil etc. if all chef does exact task than the final dish is perfectly cooked. if some chef makes mistake in adding item like adding more salt the dish becomes too salty.

In the cat versus dog neuron network each node is for shape, eyes, ears etc. that's just a really dumb down way of looking it. It's not really doing that but each node is basically looking at a certain feature in the image and then image has that the feature, the information can pass through to the next layer. If it doesn't have that feature then the information is not passed on to the next layer.

These nodes and linkages determine how much data flows through to the next layer. Ther first layer is called input layer and the last one is output layer and then all layers in between are called hidden layers.

3. How does an AI actually learn?

You can't just have any random Neuron Network and it just magically how to identify images of cats and dogs. So, first when you build a neural network the values of the nodes and layers are probably just going to be random values or they could be pre-trained values form an existing model. But how do you get it to be super good at identifying images of cats and dogs. In other words, how do to get your desired purpose.

Well, you need to feed it data. Lots and lots of data so you're going to prepare tons of images of cats and dogs and then you label it according to the photo like cat or dog. Basically, label is the answer that the AI needs to learn from the labeled input image. This is called Supervised Learning where you label the data... (2). Also, another type of learning called Unsupervised Learning where the AI need to learn to categorize data by itself without any guidance from the human.

Determining the architecture of a Neural Network is very complicated and there are infinite possibilities of how many layers you can have... (1), how many nodes in each layer you can have. Different AIs with different function have different architectures. So, they could have vastly different numbers of layers and nodes. Also keep in mind that even though neural network is the backbone of all the AI that we all know today, there are different architectures depending on the AI's purpose and function. For example, we have convolutional neuron networks or CNNs for processing images and object recognition. We have neuron networks or RNNs as well as LSTMS or long- and short-term neural networks and these are often used for forecasting time series or predicting for example the stock market. We also have the Transformers architecture which is used by most of the major large language models that we know today including GPT, cloud, LLaMA etc..... (4)(5).

4. How dose ChatGPT works?

So again, it is a kind of a same thing, it is training a neural network but, in this case, instead of images of cats or dogs we train it on a language and all of the data in the world.

Of course, the neural network of ChatGPT is way more complicated than this. rumors claim that GPT-4 has 1.76 trillion parameters. Now, I am giving a simple example of how they train it but again I am oversimplifying a lot so you can get a high-level understanding of it there are a lot of details that I have left out. So, for example you could feed it data like this:

Which planet has the most moons? - Saturn.

What country has won the most World Cups? – Brazil.

These are very basic questions and you can see how complex it can get if you give it a prompt like write an essay on XYZ or does creatine help build muscle? and then it spits out an answer like creatine supplementation generally enhances muscle strength increases fat free mass, etc.

This is a very long form and complicated answer. So, how does it know if it got that answer right or wrong it's not as simple as identifying an image and determining if it's a cat or a dog and that's why initially how open AI trained. GPT had lots of humans manually verify its answers to determine if GPT got it right or wrong. This is called reinforcement learning from human feedback also known as RLHF... (5) and if it gets the answer wrong for example if for this question which planet has the most moons it answered Jupiter instead of Saturn. Then it would get a penalty for it and then through gradient descent it would tweak these knobs and dials further until the entire network gets all the answers correctly no matter what prompt you give it... (6). In essence, that's how these large language models work.

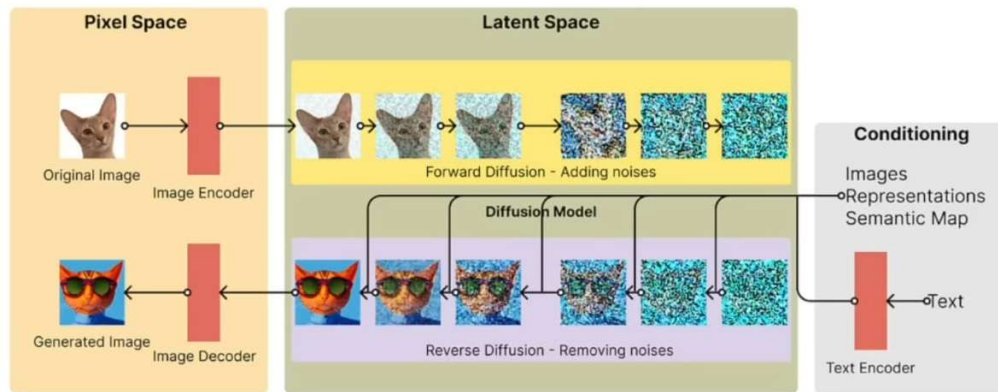
5. Why are some models better than others?

For example, why is cloud 3 better than GPT 3. That's because cloud 3 has a lot more parameters so that either means more layers more nodes in each layer. Generally speaking, the more complex the neural network, the better it is at handling complex tasks... (2)(3). That's why computing and AI chips are in such high demand there's now a lot of investments flowing into AI chip companies because they see the potential of huge growth in the space in the upcoming years

For example, Nvidia's flagship H-100 GPU is also in such high demand in fact it was sold out for all of 2023 this is like the most prized commodity in the tech space and you can see the major tech companies like Microsoft beta they have purchased an estimated 150 thousand of these H-100 GPUs to power their computing.

6. How does generation work?

Now that you know how a neural network is trained you can probably guess how in the generation works as well. Instead of feeding its images of cats or dogs you would feed it a lot of images with a text description and again you just feed millions of these images each with a labeled text description to the neural network that eventually gets good at producing an image based on a text description or a prompt... (7) now here's how stable diffusion works:



You can see that the neural network doesn't actually generate an image it removes noise in sequential steps to eventually your desired image so it's not starting from a blank canvas it's actually starting from random noise and then in each step it removes some noise until you get your generated image so this process is called reverse diffusion... (8). Now to train it, what this actually does in the back end is you feed it original image and then in each sequential step it actually adds noise to the image in a process called forward diffusion until it reaches an image of just noise and then through this process of forward diffusion and reverse diffusion it's able to eventually learn how to generate an image based on a prompt.

7. Is AI actually copying or stealing art?

I know a decent portion of the artist community. I'm not saying all of them but a decent amount of them is quite hostile towards AI. They really hate it and they think that AI is stealing their art, stealing their jobs etc.

When a neural network from for example mid journey or stable diffusion is trained on image data. It might be given something like Greg Rutkowski style or maybe Ghibli style or anime style. Once the AI learns to associate one particular image style with the word Ghibli or anime or one image with the word Greg Rutkowski style it would produce images in that style if you give it that prompt. But is this really copying or stealing? Essentially artists hating this.

We humans also draw images in these styles and we also have fan art (Humans draw artwork that are based on original content from other artists). So, why aren't artists hating on people who are producing fan art based on some other original content but they're hating on AI which is essentially doing the same thing. It's just learning through this brain to associate a particular style and then reproducing that style. It's not copying and pasting the exact picture. It's just learning a style just like a human brain would learn a particular style of image.

This also brings up the concern about AI allegedly stealing content from publishers like the New York Times which is now suing open AI for copying their content... (14) but again is this argument really valid?

The New York Times

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

At the end of the day, they are suing neural network which is trained on all the data in the world. This is just an artificial brain that you can say has learned information from the Internet and from the world. So yes, it could have been fed a New York Times article and learned information from it but it's not really plagiarizing, it's not copying and pasting a New York Times article for work.

Let's talk about a New York Times article claiming that this woman Mira Murati fired Sam Altman which is totally incorrect. By the way, after this original New York Times article came out if other publishers also published the same content such as Business Insider and New York Post, they all just copied this original New York Times article. So, is this plagiarizing? They're all producing secondary content based on this primary source so why isn't New York Times suing Business Insider and New York Post or all other publishers that are creating content by copying the New York Times. But they're suing the neural network.

Again, this is just a digital brain. One can say that it's taking information from the Internet. It could include New York Times articles and then learning from that information just like we humans would. Then rewriting that information when we're prompted to do so.

Neural network is great at predicting patterns in life. There are certain patterns on what makes a good essay there are certain patterns, on what is considered a dog, there are certain patterns on what is considered a water-colored painting or a Ghibli style image.

“Life is full of patterns”

The best salespeople follow similar playbooks the best businesses follow similar strategies the best YouTube videos also use the same strategies over and over again life is full of patterns and the neural networks job is to identify these patterns and reproduce them.

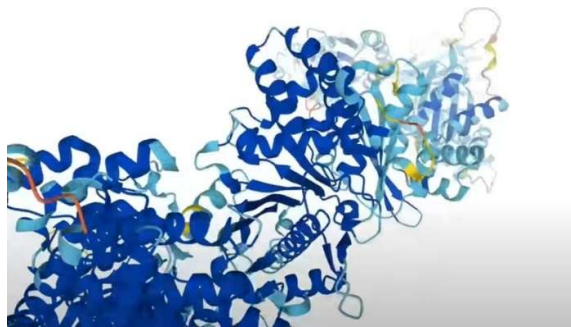
8. Can AI solve unsolvable math problems?

In the starting, we talk about a roomer of a leaked document which claims to be about the mysterious Q star project that open AI is working on now this is a very controversial document because it claims that they trained in AI that was able to break encryption systems.

Encryption is what secures literally the whole world digitally from our passwords or credit cards government data the stock market wireless networks etc.... (11). if an AI is able to break this system, then the as we know it could collapse instantly. Now, there's no way an AI could break encryption because there's no formula for you to easily find the answer or find the password. Once you have the password you can easily determine that "it's correct", but the reverse is not true there's no fixed way to guess did password besides brute force and for these advanced encryption systems using brute force guessing that means guessing all the possible combinations of letters to get that password. It's going to take a very long time. So, there's no way that AI could break encryption... (12).

AI can approximate any function or pattern life is full of patterns but many patterns cannot be explained by a simple formula. for example, protein synthesis. How certain protein molecules interact with one another and fold into these complex 3D structures is just something we cannot mathematically map out with a formula. It's just too complex.

"Protein folding presents a problem called Leventhal's paradox... (10). Which states that proteins can potentially adopt an astronomical number of confirmations or shapes due to the flexibility of their peptide bonds. Leventhal estimated that even a small protein of 100 amino acids could sample 10 to the power of 300 possible confirmation. if we were to brute force against the correct shape well there are 10 to the power of 300 possible shapes we could guess. Which would take any eternity to get right.



However, proteins typically fold into their native structure within milliseconds to seconds which is much faster than the time scale predicted by the sequential search of all possible confirmations.”

- ChatGPT-4 (AI)

Now for decades, scientists were not able to find a mathematical formula to figure this out however finally Alpha Folds from Google DeepMind was able to solve this problem... (9). Using AI and deep learning they were able to predict the very high accuracy how any amino acid definition of amino acids would fold together to form a 3D structure.

So, can AI solve unsolvable math problems? As long as there is an underlying pattern behind that problem which we may or may not be aware of right now, it could very well solve that problem.

9. Can AI beat humans at anything and everything?

As I've shown you the neural network is basically a brain. Brain is also a series of knobs and switches which are interconnected into this network specifically the human brain has 86 billion neurons but structure is the same thing as neural network. So, what if we built an AI or a neural network that exceeds 86 billion neurons. If it's built the same way in theory, it could very well outcompete humans at almost everything. The more complex the network or the more neurons in the network in theory - the smarter it is. again, life is full of patterns and AI is all about pattern recognition... (15). There are patterns in psychology, human psychology is very predictable. Medical diagnosis is also just pattern recognition. how to create a successful business or how to make money in life or how to be successful in life it's the same playbook over and over, we're not inventing anything new here and since AI is so good at pattern recognition it can in theory eventually be better than us or already is better than us in these tasks.

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